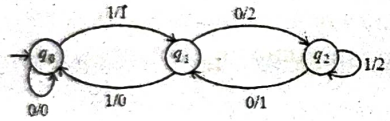
BANSAL GROUP OF INSTITUTES
BHOPAL INDORE MANDIDEEP

SushilaDeviBansalCollege
An Institute of Bansal Group of Institutes
Umaria,A.B.Road,NearRau,Indore -453331

DATE: 08/09/2026
DEPARTMENT: COMPUTER SCIENCE & ENGINEERING
B.TECH: 3RD YEAR 5TH SEMESTER
SUBJECT CODE: CS-501
SUBJECT NAME: THEORY OF COMPUTATION
IMPORTANT QUESTIONS (AS PER RGPV PATTERN)

OK. Good

Unit 1:

Q. No.	Question	CO	Learning Level	RGPV Month & Year
1	Define Finite Automata. Differentiate between a Deterministic Finite Automata (DFA) and a Non-Deterministic Finite Automata (N-DFA) with suitable examples.	CO1	L3	May/Dec 2025
2	Differentiate Mealy machine & Moore machine with diagram.	CO1	L2	June 2021
3	Construct Moore machine for the following Mealy machine. 	CO1	L3	Dec 2023
4	Design a Mealy machine that output 1 whenever the input symbol is '1' otherwise output 0.	CO1	L3	Dec 2025
5	What is a composite machine? Explain how finite automata act as language acceptors and translators.	CO1	L2/L3	May 2023

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Unit 2:

Q. No.	Question	CO	Learning Level	RGPV Month & Year
1	State and prove Arden's Theorem. How is it used to find the regular expression corresponding to a given Finite Automaton? Explain Arden's theorem. R=Q+RP => R=QP*	CO2	L2	Dec 2025
2	For Σ = {a, b}, construct DFA's that accept the sets consisting of i) All the strings with exactly one 'a'. ii) All the string with atleast one 'a'. iii) All strings with atleast one 'a' and followed by exactly two b's.	CO2	L3	May 2023
3	Describe the NFA and DFA in brief. Also, explain the construction of DFA for the following regular expression : r = (0+1)*(00+11) (0+1)*	CO2	L2/L3	Nov 2023 Dec 2025
4	Describe the following sets by regular expression. i) Strings containing 1100 as substrings. ii) The set of all strings of 0's are 1's ending with 00. .....}. iii) {Λ, a, aa, aaa, aaaa, ... ii) Write a regular expression for the language over {a,b}that contains string ending with'ab'.	CO2	L3	Dec 2025 Dec 2024 May 2023
5	Define two way finite automata? Also Define DFA & it's Applications.	CO2	L1/L2	Dec 2020 June 2020

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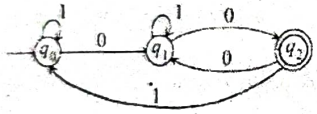
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Unit 3:

Q. No.	Question	CO	Learning Level	RGPV Month & Year
1	a) Show that the following grammar is ambiguous. S→aSbS/bSaS/∈ b) What is meant by ambiguous grammar? Test whether the grammar is ambiguous or not. S→ SS/a/b	CO3	L4	June/Dec 2020 Nov 2023
2	Convert the grammar S → AB, A → BS/b, B → SA/a Greibach normal form.	CO3	L3	Dec 2024
3	Derive the regular expression from this finite automata. 	CO3	L4	Nov 2023
4	Explain the Chomsky Hierarchy of Grammars in detail, detailing Types 0, 1, 2, and 3 along with their recognizing automata.	CO3	L4	Nov 2022
5	Convert grammar S→aSb|SS|∈ to Chomsky Normal Form (CNF).	CO3	L3	June 2025
6	What is ambiguity in Context-Free Grammar (CFG)? How do you prove whether a given grammar is ambiguous or unambiguous?	CO3	L4	Dec 2024

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Unit 4:

Q. No.	Question	CO	Learning Level	RGPV Month & Year
1	Give An NPDA that simulates the following grammar. S → aBB | cDD B → cD | aS D → dD | d	CO4	L4	Nov 2022
2	Construct a Pushdown Automaton (PDA) for accepting the language $L = \{w C w^R \mid w \in \{0, 1\}^*\}$ or $L = \{a^n b^n \mid n \ge 1\}$.	CO4	L3	Dec 2025
3	Is NPDA or DPDA equivalent or not? Illustrate with an example.	CO4	L3	Dec 2024
4	Design a Push down automata. i) anbn, n ≥ 1 ii) wwR, w ∈ (a, b)2	CO4	L4	May 2023
5	Describe the conversion from PDA to CFG using the standard algotrithm.	CO4	L2	June 2025

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Unit 5:

Q. No.	Question	CO	Learning Level	RGPV Month & Year
1	Explain NP problem in detail with example.	CO5	L2/L3	Dec 2024
2	Write Short not on: a) Turing Machine with their techniques for construction b) Halting Problem	CO5	L2	Dec 2025 May 2023
3	What is decidable and undecidable problems? Explain with Example.	CO5	L2	Dec 2025 Dec 2020
4	Explain Recursively enumerable language.	CO5	L2/L3	Dec 2025 May 2023
5	Write short note on a)Universal Turing machine b) Post corespondance problem	CO5	L2/L4	Dec 2024,

ame of Course Coordinator : Ms. Nidhi Kashive **HOD SDBCE** HOD, SDBCT 03/03/24